

Finite-Regulator Source Insertions and Obstruction Tests

in a Gauge-Covariant $SU(2)$ Renormalization Step

An exploratory research handoff

YesterdaysLemon

assembled from an AI-assisted open research ledger

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Status. This paper is not a solution of the Yang–Mills existence and mass gap problem. It proves no continuum limit, no infinite-volume quantum field theory, and no mass gap. It reports finite-regulator algebra, conditional one-step constructions, and obstruction tests from an exploratory public repository. The repository-level evidence status is E0; the strongest individual auxiliary claims are internally checked at E2 and have no independent human review.

Abstract

We record the outcome of an exploratory attempt to insert a local plaquette-energy source into a gauge-covariant, Balaban-type renormalization step for four-dimensional $SU(2)$ Wilson lattice gauge theory. On one selected finite-regulator small-field branch, the work derives exact source-jet identities, transports one marked plaquette through the finite RG-II hard-core and cluster algebra, and isolates explicit conditional hypotheses under which completed external-field derivatives would be uniformly local. Several natural repairs of the remaining physical- U pullback fail: independent-link complex collars shrink like the square of the lattice scale; diagonal scale balancing restores the same loss through Cauchy estimates; and separate absolute values destroy possible within-column U/J Ward cancellation.

The final checkpoint has two parts. First, differentiation of the variational background equations gives an exact affine range description and elliptic factorization for the source kernel. Second, a source-normalized finite periodic Cartan model has an exact cochain identity. A localized coarse bond has nonzero coarse curl, and discrete Stokes forces raw s^{-1} and gradient/curl s^{-2} lower bounds for every right inverse in that model. The retained primary sources do *not* establish that this finite model embeds into their full multiscale H_1 domain or Hessian-selected range. That embedding, completed weak-dual control, nonzero-source polymer analyticity, large fields, iteration, regulator removal, reconstruction, and the infrared mass gap all remain open. The purpose of this paper is to preserve the exact discoveries and failure modes in a form suitable for future researchers and future language-model agents.

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1 Status, target, and interpretation

1.1 The actual problem

The Clay problem asks, for every compact simple gauge group G , for a nontrivial pure Yang–Mills quantum field theory on four-dimensional space, axiomatic control at least at the level described by Jaffe and Witten, the required short-distance Yang–Mills behavior, and a reconstructed Hamiltonian whose spectrum has a finite positive gap above the vacuum [1]. A finite lattice, a formal path integral, perturbation theory, numerical glueball data, or a fixed-coupling strong-coupling gap is not this theorem.

The working scaffold in the repository is

$$\begin{aligned} &\text{Wilson regulator} \longrightarrow \text{gauge-covariant RG} \longrightarrow \text{local observables} \\ &\longrightarrow \text{OS reconstruction} \longrightarrow \text{infrared bridge} \longrightarrow \text{spectral gap.} \end{aligned} \quad (1.1)$$

Wilson’s compact regulator supplies exact finite-dimensional gauge invariance [2]. Osterwalder–Schrader reconstruction, Osterwalder–Seiler lattice positivity, and Lüscher’s transfer matrix describe the desired positivity and spectral scaffolding [3, 4, 5, 6]. Balaban’s papers provide the principal ultraviolet RG input used here [7, 8, 9, 10, 11, 12, 13].

These inputs do not supply the arrows in (1.1). In particular, if a is the lattice spacing and T_a has leading spectral values λ_0, λ_1 , then

$$m_{\text{lat}}(a) = -\log \frac{\lambda_1}{\lambda_0}, \quad \Delta_{\text{phys}}(a) = \frac{m_{\text{lat}}(a)}{a}. \quad (1.2)$$

A finite nonzero continuum gap would require

$$m_{\text{lat}}(a) = a\Delta + o(a). \quad (1.3)$$

A fixed positive gap in lattice units normally corresponds to a correlation length of order one lattice spacing and does not establish (1.3).

1.2 Evidence vocabulary

The public ledger [yang-mills-research](#) separates project status from the status of an individual auxiliary statement. Its current interpretation is:

- **E0 (project):** exploratory; no candidate solution wording is permitted.
- **E1 (individual):** a derivation or source transcription with incomplete independent checks.
- **E2 (individual):** exact finite algebra or a primary-source consequence with executable regressions and explicit objections.

No E2 label certifies deep mathematical truth. The tests check identities, finite countermodels, links, and governance boundaries. No independent human expert has reviewed the chain.

1.3 Scope of this report

The developed material concerns:

- finite four-dimensional periodic regulators;
- primarily the group $SU(2)$;
- one gauge-covariant block step;
- a selected near-identity small-field branch;
- one localized plaquette mark and predominantly the first source derivative at $t = 0$;
- conditional external- (U, J) analytic tubes.

It does not identify the selected coordinate law with the unrestricted raw block transform. It does not construct a nonzero-source polymer disk, treat the complete large-field operation, iterate the RG, remove either regulator, restore all Euclidean symmetries, reconstruct a continuum Hilbert space, or prove infrared decay.

2 Finite source identities and the selected branch

2.1 Finite positive kernels

Let (Ω, ν) be a finite measure space, let $w > 0$ be integrable, and let O be a bounded real observable. Write

$$Z(t) = \int_{\Omega} e^{-tO(\omega)} w(\omega) d\nu(\omega), \quad \mathcal{Z}(t) = \frac{Z(t)}{Z(0)}. \quad (2.1)$$

The following basic statement is independent of gauge theory.

Proposition 2.1 (Exact finite source jets). *For (2.1),*

$$\partial_t \log Z(t)|_{t=0} = -\mathbb{E}(O), \quad \partial_t^2 \log Z(t)|_{t=0} = \text{Var}(O). \quad (2.2)$$

Mixed derivatives give covariances. The normalized ratio $\mathcal{Z}$ is unchanged by any positive, source-independent rescaling of w . Furthermore, a centered bounded observable gives an explicit zero-free scalar disk; one convenient bound used in the ledger is

$$|z| < \frac{2 \log 2}{\text{diam } O}. \quad (2.3)$$

This proposition gives exact finite-volume cumulants and projective invariance. It does not give a polymer expansion at $t \neq 0$, uniformity in a regulator, or a continuum measure.

2.2 Wilson fields and a local mark

On a finite torus Λ , an $SU(2)$ bond field assigns $U(b) \in SU(2)$ to each oriented nearest-neighbor bond, with $U(-b) = U(b)^{-1}$. For an oriented plaquette p , let $U(p)$ be the ordered product around its boundary. A representative local plaquette-energy insertion is

$$O_p(U) = 1 - \frac{1}{2} \text{Re Tr } U(p). \quad (2.4)$$

The Wilson action is a sum of such terms. A profile source would replace the action by $S(U) + \sum_p t_p O_p(U)$.

The research program asks for a one-step formula in which derivatives in t_p are represented by gauge-invariant, exponentially localized coarse operators and polymer remainders with volume-independent constants. The first source derivative is represented algebraically by a dual number ϵ , $\epsilon^2 = 0$. This keeps exactly one marked occurrence through products without pretending that a common nonzero source disk has already been proved.

2.3 Selected coordinate law

The Balaban variational construction provides a local analytic minimizing background on an admitted small-field patch [11]. The repository combines that construction with the RG-I chart covariance [12, 11], fixes one near-identity branch, retains the chosen gauge chart and cutoff, and normalizes the resulting finite integrand to a probability law. Within one relatively compact patch, analytic tests and scalar source ratios are holomorphic, and real coarse gauge transformations act by push-forward.

Remark 2.2 (The selected/raw distinction). The selected branch is an operational coordinate law. No theorem in the ledger identifies it with the unrestricted group-delta block transform. All source results below inherit this restriction unless explicitly stated otherwise.

3 One marked plaquette through the finite RG algebra

3.1 Distinguished-vertex Ursell algebra

Consider a finite hard-core polymer gas with activities $z(X)$, and add a single marked activity $\dot{z}_p(X)\epsilon$. Differentiating the connected logarithm selects exactly one distinguished occurrence. The ledger checks the connected-graph/Ursell coefficient including repeated labels and the $1/n!$ convention. Non-root components have zero marked derivative, so the derivative of a product selects the unique component containing the mark.

This is elementary but important: expanding a derivative over all later placements manufactures terms that are absent from the exact dual-number product. The mark must be routed through the actual factorization before absolute values are taken.

3.2 Conditioning and the whole-integrand contour

The marked weakening integrand contains conditioning factors, Gaussian relative-field factors, cut-offs, and later scale reorganizations. The repository uses one Cauchy contour for the *whole* integrand.

Separate majorants for separately differentiated factors do not bound the product unless a common holomorphic domain and a joint integrable majorant are available.

The source-faithful routing calculation places the local mark on the conditional interior field B , not on a shifted variable chosen only for convenience. For the relative Gaussian factor, the mark is independent of the relative variable, so its additional Gaussian moment cost is exactly zero. This closes a finite-term routing issue; it is not a proof of a nonzero-source gas.

3.3 Marked resummation and a fixed-gas first jet

After the Section-2 reorganization in Balaban's RG-II cluster construction [13], the ledger obtains one decorated marked activity and a unique marked-component factorization. Under its explicit smallness hierarchy, doubled marked-amplitude refinement, and a separate ordinary Kotecký–Preiss ceiling [14], the marked seed can be resummed with positive bounds.

In the notation of the ledger, a representative bound is

$$|W_p^{\text{post}}(Z)| \leq 4K_{\text{lift}}\alpha_6^{-1}\mathcal{B}_\bullet \exp\left[-(1-8\delta)\frac{L}{2}\kappa d_{k+1}(Z)\right]. \quad (3.1)$$

After the designated animal-entropy spend, the rooted exponent is

$$(1-9\delta)\frac{L}{2}\kappa. \quad (3.2)$$

The final source hierarchy chooses $(1-10\delta)(L/2)\kappa = \kappa$, leaving the declared output exponent.

Theorem 3.1 (Conditional fixed-regulator marked first jet). *Assume the complete fixed-gas source hierarchy recorded in claims YM-RG-020–YM-RG-027, including the separate ordinary pinned ceiling, the doubled marked-amplitude condition, periodic compatibility $LM \mid N$, and a nonempty common shifted-branch domain. Then one interior plaquette mark passes exactly through the finite RG-II algebra. Its final connected derivative at $t = 0$ is absolutely summable with the output polymer exponent κ , and completed shifted-branch numerator/denominator pairs represent the same projective first jet without an orbit factor.*

Boundary of Theorem 3.1. It is a first derivative at $t = 0$ in independent external coordinates. It neither constructs source-dependent activities on a common t -disk nor converts the derivative to the physical minimizing coarse field.

3.4 Geometry, animals, and completed support

For the declared closed-cube model, the literal union of a connected tuple is the least admitted hull. A contained-tree metric and rooted-animal count give an explicit sufficient entropy window. In four dimensions the convenient threshold appearing in the ledger is

$$\eta_{\text{animal}} > 64 \log 8. \quad (3.3)$$

The final ordinary RG-II gas is mapped into this abstract bound only after keeping its one-sided metric comparison and species aggregation explicit.

The shifted-branch construction is synchronized over dual numbers. Periodic covering-space endpoint geometry removes an artificial tree-lift hypothesis. Because the completed coefficient is

grouped by its literal union and the Ursell and normalization factors are field-independent, the external- (U, J) dependence is confined to bonds whose geometric segments meet the interior of that union. On the aligned integer-wall cubulations, the completed J -support has zero endpoint halo.

4 Conditional physical pullbacks

4.1 The affine J tube

Let $\widehat{\mathcal{C}}_p^+(s, R; U, J)$ denote a completed connected coefficient. The compatibility hypothesis (H_J^{conn}) requires one raw affine J representative whose restrictions agree across all overlapping charts on each complete shifted branch. Under that hypothesis, the existing marked majorant can be rerun in a coefficientwise local H^∞ norm:

$$\sum_{s,R} e^{\kappa d_{k+1,s}(R)} \sup_{\|J-\bar{J}\|_\infty < \Delta_J} |\widehat{\mathcal{C}}_p^+(s, R; U, J)| \leq B_{\text{conn}}. \quad (4.1)$$

Banach-line Cauchy and the exact support theorem then give

$$\sum_{s,R} e^{\kappa d_{k+1,s}(R)} \|D_J \widehat{\mathcal{C}}_p^+(s, R)\| \leq \frac{B_{\text{conn}}}{\Delta_J}, \quad (4.2)$$

with no bond-volume or branch factor.

The hypothesis (H_J^{conn}) is not proved by the source papers. Equation (4.2) is therefore a conditional completion theorem, not an unconditional physical source derivative.

4.2 A product U/J tube

Raw independent-link U balls are poorly adapted to the curvature domain. The ledger instead introduces a relative-log, background-covariant norm controlling raw values, covariant gradients, and transported plaquette curls. Under the real-center hypothesis (H_{rc}), every retained physical center has one real/unitary representative with a uniform curvature margin. Ordered matrix estimates then give a positive product U/J tube and full dual derivative norms.

Neither the primary-source audit nor the finite tests prove (H_{rc}) for the entire completed minimizing family. This is one of the main unresolved uniformity assumptions.

4.3 Support-anchored physical J

Let $S_{p,s,R}$ be the complete active external- J support of a coefficient. After composing the completed J -derivative with the audited source kernel, the repository obtains, under its common kernel and chart constants,

$$\begin{aligned} & \sum_{s,R} e^{\kappa d_{k+1,s}(R)} \sum_{y'} \mu(y') e^{\gamma d_B(S_{p,s,R}, y')} |\mathcal{L}_{p,s,R}^{J,\text{conn}}(y')| \\ & \leq \frac{C_\chi c_1(\alpha_\gamma) \bar{E}_J^{\text{conn}} B_{\text{conn}}}{\Delta_{\text{tube}}}, \quad 0 \leq \gamma < \frac{\delta_0}{8}. \end{aligned} \quad (4.3)$$

The weight is anchored at the *full active support*, not automatically at the marked plaquette. A stronger plaquette-rooted theorem needs an additional endpoint or coefficient-weighted hypothesis.

4.4 Physical U , quotient synthesis, and transported curl

For an active bond set I , directions invisible to the coefficient form a nullspace N_I . The supported derivative naturally lives in the dual of the quotient X_U/N_I , not in raw coordinate ℓ^1 . Exact finite-dimensional phase duality reduces the conditional physical- U estimate to a quotient-synthesis operator norm for the converted source columns.

The covariant curl is not an independent mystery row. Under the real-center curvature margin, the ledger proves the exact four-link identity

$$\mathcal{D}_{\bar{U}}^\xi a = D_\mu^\xi a_\nu - D_\nu^\xi a_\mu + \xi^{-1}(I - \text{Ad}_{d\bar{U}(p)})(a_\nu + T_\nu a_\mu). \quad (4.4)$$

Consequently, the raw/gradient/operator/curl norm is uniformly equivalent to a raw-plus-covariant-gradient norm on that real-center tube. A Lipschitz cutoff of fixed physical width ρ yields

$$C_{\text{ext}}(\rho) \leq C_{\text{eq}} \left(1 + \frac{c_{\text{Ad}}^{\text{RG}}}{\rho} \right). \quad (4.5)$$

A flat longitudinal path proves that a fixed number m of lattice layers necessarily costs order $1/(m\xi)$. Thus a fixed-layer collar cannot replace the physical-width collar uniformly as $\xi \rightarrow 0$.

5 Why the obvious strong-norm repairs fail

5.1 Independent-link collars

On the finest homogeneous patch, a complex perturbation concentrated on one bond changes an adjacent plaquette at first order. The small-field curvature domain allows only $O(\xi^2)$ deviation in the relevant coordinates. Therefore a full complex independent-link ball contained in that domain has radius at most $O(\xi^2)$. Generic Cauchy estimates on such a ball carry an $O(\xi^{-2})$ loss.

This is a finite-regulator analytic-domain obstruction. It does not say that every structured source direction is bad.

5.2 Diagonal scale balancing

The source kernel bounds in the variational paper contain raw and gradient output powers of the schematic form s_j^{-1} and s_j^{-2} [11]. Weighting the chart norm by s_j and s_j^2 cancels those powers algebraically. However, the same weights make the full chart ball's admissible radius $O(\xi^2)$ on the finest patch. Banach-line Cauchy returns the $O(\xi^{-2})$ loss.

Proposition 5.1 (Scale-renorming no-go). *A diagonal raw/gradient weighting that cancels the printed s_j^{-1}, s_j^{-2} row powers does not, by itself, produce a regulator-uniform physical- U derivative theorem through generic Cauchy estimates. The analytic radius collapses with the same scale power that the row weighting removes.*

A separate flat curl-free longitudinal example shows that retaining an unweighted curl component does not evade the gradient-domain obstruction. Proposition 5.1 does not rule out a range-restricted tube, a direct weak-dual estimate, or structural cancellation.

5.3 The joint U/J Ward quotient

A physical coarse-source column changes both the relative U field and the auxiliary J coordinate. Write

$$V_\alpha = (A_\alpha, L_{\bar{U}} A_\alpha). \quad (5.1)$$

The two summands must be paired before absolute values. A one-dimensional bidisc example in the ledger shows that separate absolute values can destroy an exact Ward cancellation.

Let H_F contain coefficient-inactive directions and simultaneous complexified gauge tangents. Conditional on the completed coefficientwise Ward identity, the sharp coefficient-independent first-jet constant is

$$\|Q_{H_F} K_F^w\|_{\ell^\infty(\text{sources}) \rightarrow E_\xi/H_F}. \quad (5.2)$$

Arbitrary phases in the ℓ^∞ source ball forbid relying on cancellation between distinct source columns. Genuine cancellation must occur within each joint U/J column or in the quotient.

The Ward premise and a regulator-uniform bound for the actual converted synthesis are not proved. A bounded gauge-invariant Cartan plaquette functional gives a logical same-cell transverse countermodel under the retained abstract hypotheses, but that countermodel is not an actual Balaban polymer coefficient.

6 Exact elliptic factorization and a finite-model Stokes obstruction

6.1 Measure-normalized columns

Let c be a physical coarse-source coordinate with source measure μ_c . Use the density convention

$$D_B \mathcal{H}[h] = \sum_c \mu_c A_c h_c. \quad (6.1)$$

The coordinate column constrained by differentiation is

$$\kappa_c = \mu_c A_c. \quad (6.2)$$

At a fixed source center $\bar{B}$, define the differentiated averaging and projected-Landau operators $C_{\bar{B}}$ and $S_{\bar{B}}$. Under the repository density convention (6.1), differentiating variational-paper Eqs. (20)–(21) gives [11]

$$C_{\bar{B}} \kappa_c = e_c, \quad S_{\bar{B}} \kappa_c = 0. \quad (6.3)$$

Applying (6.3) to the density A_c instead of the coordinate column κ_c loses a source-measure factor.

Proposition 6.1 (Affine constrained range). *Let $C : X \rightarrow Y$ and $S : X \rightarrow Z$ be maps between finite-dimensional Hilbert spaces, and assume $C|_{\ker S}$ is onto. If $Z = \ker C \cap \ker S$ and $R_C : Y \rightarrow \ker S$ is any right inverse, then all solutions of $C\kappa = e$, $S\kappa = 0$ are*

$$\kappa = R_C e + Z. \quad (6.4)$$

A detector $\ell \in X^*$ is fixed by these constraints exactly when

$$\ell|_Z = 0 \iff \ell \in \text{ran } C^* + \text{ran } S^*. \quad (6.5)$$

Thus averaging and projected Landau select an affine slice, not a unique column. The variational Hessian supplies the additional selection.

6.2 The exact differentiated factorization

The variational source uses three different constrained lifts [11]: H is the original lift used in the nonlinear restoration $HD(A')$, H_1 is the improved lift in Eqs. (174)–(177), and H_0 is the alternative lift in Eqs. (179)–(190). They must not be identified with one another or with the later Landau-to-axial gauge map. Put

$$A_B = \mathcal{A}_0(B) + H_0 B, \quad T_B = D_B A_B, \quad W_B = V''(A_B). \quad (6.6)$$

With $\tilde{G} = (\Delta_a - \Delta^{(2)})^{-1}$, the differentiated equation in the variational paper is

$$\frac{\delta \mathcal{A}_0}{\delta B} + \tilde{G} W_B T_B = \tilde{G} \Delta^{(2)} H_0. \quad (6.7)$$

Since $\delta_B \mathcal{A}_0 = T_B - H_0$, multiplication by $\tilde{G}^{-1}$ gives

$$(\Delta_a - \Delta^{(2)} + W_B) T_B = \Delta_a H_0. \quad (6.8)$$

The background derivative is therefore

$$\boxed{K_B = D_B \mathcal{H}(B) = (I - HD'(A_B))(\Delta_a - \Delta^{(2)} + V''(A_B))^{-1} \Delta_a H_0.} \quad (6.9)$$

Equation (129) of the source, together with $G = \Delta_a^{-1}$, implies

$$\Delta_a H_0 = Q^*(QGQ^*)^{-1}(L^{j(\cdot)}\eta)^{-1}. \quad (6.10)$$

The order- -2 inverse in (6.9) is fed by an order- $+2$ source. It cannot be read in isolation as a two-power smoothing theorem. The printed expansion also yields the exact consequence

$$K_0 = D_B \mathcal{H}(0) = H_1. \quad (6.11)$$

Equations (6.8) and (6.9) are derived from the printed differentiated Eqs. (182)–(183); (6.10) is derived from printed Eq. (129) and the preceding definition $G = \Delta_a^{-1}$; and (6.11) is the derivative consequence of the printed expansion (177) [11]. The Stokes calculation below has a deliberately weaker status.

6.3 A source-normalized finite periodic model

Take

$$\Lambda_f = (\mathbb{Z}/4\mathbb{Z})^2 \times (\mathbb{Z}/2\mathbb{Z})^2, \quad \Lambda_c = (\mathbb{Z}/2\mathbb{Z})^2 \times (\mathbb{Z}/1\mathbb{Z})^2, \quad (6.12)$$

and set $L = 2$. Restrict to a fixed Cartan generator $T \in \mathfrak{su}(2)$ and fields constant in the last two coordinates. Write $a = \eta\kappa$. The finite model retains the flat-Cartan Q_0, Q'_0 and projected-Landau normalizations:

$$\begin{aligned} (Ca)_1(X, Y) &= \frac{1}{4} \sum_{r \in \{0,1\}^2} \sum_{t=0}^1 a_1(2(X, Y) + r + te_1), \\ (Ca)_2(X, Y) &= \frac{1}{4} \sum_{r \in \{0,1\}^2} \sum_{t=0}^1 a_2(2(X, Y) + r + te_2). \end{aligned} \quad (6.13)$$

The scalar block map is the full four-dimensional mean; on invariant fields it restricts to the 2×2 mean. The model Landau projector is the orthogonal projection onto $\Delta \ker P_4$.

Remark 6.2 (Source boundary). The averaging and gauge-fixing papers support the normalizations used in (6.13) [8, 9]. They do not, in the material audited here, prove that the anisotropic finite periodic model (6.12) is an embedded patch of the complete multiscale H_1 construction, its full Q' nullspace, or its boundary conditions.

6.4 Exact representative and discrete Stokes

Let e_c be one positive coarse 1-bond at $(X, Y) = (0, 0)$. The following rational field is extended constantly in the spectator coordinates, with $a_2 = a_3 = a_4 = 0$:

$$a_1 = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} & \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} & \frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (6.14)$$

Direct substitution gives $Ca = e_c$. If $g = d^*a$, then

$$g = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (6.15)$$

The field Δg is constant on each full 2^4 block. Hence $\Delta g \in \text{ran } P_4^* = (\ker P_4)^\perp$, so

$$g \perp \Delta \ker P_4, \quad Rd^*a = 0. \quad (6.16)$$

Thus the model's localized right-inverse/Landau slice is nonempty.

The bond map is a cochain map:

$$d_c C = C_2 d, \quad (6.17)$$

where a selected row of C has ℓ^1 -mass 2 and a selected row of C_2 has ℓ^1 -mass 4. Since $(d_c e_c)(0, 0) = 1$, every model field with $Ca = e_c$ obeys

$$\|a\|_\infty \geq \frac{1}{2}, \quad \|da\|_\infty \geq \frac{1}{4}, \quad \|\delta a\|_\infty \geq \frac{1}{8}. \quad (6.18)$$

With $s = 2\eta$ and $\kappa = \eta^{-1}a$, this becomes:

Theorem 6.3 (Finite-model coarse-curl obstruction). *Every right inverse of the localized source in the finite model (6.12)–(6.13), with or without the projected-Landau condition, satisfies*

$$\boxed{\|\kappa\|_\infty \geq s^{-1}, \quad \|\nabla^\eta \kappa\|_\infty \geq \frac{1}{2}s^{-2}, \quad \|d^\eta \kappa\|_\infty \geq s^{-2}.} \quad (6.19)$$

The representative (6.14) has matching powers:

$$\|\kappa\|_\infty \leq \frac{3}{2}s^{-1}, \quad \|\nabla^\eta \kappa\|_\infty, \|d^\eta \kappa\|_\infty \leq 3s^{-2}. \quad (6.20)$$

Theorem 6.3 is an exact rational finite-dimensional theorem. It proves that right-inverse normalization and projected-Landau orthogonality alone do not force a two-power strong-norm gain in this model. It does *not* prove that the actual H_1 column contains this model, that the full Hessian selector retains the curl mode, or that a completed physical coefficient detects it.

7 Research handoff

7.1 The next decision, in order

The most informative next work is not another generic strong-norm estimate. It is the following decision tree.

1. **Embedding test.** Prove or disprove that the homogeneous Cartan finite model can be embedded into the full multiscale H_1 domain with its complete Q' nullspace, domain decomposition, boundary conditions, and Hessian selection.
2. **If the curl survives, test the completed weak dual.** For each completed coefficient, evaluate the actual within-column functional

$$D_U \hat{\mathcal{C}}[A_\alpha] + D_J \hat{\mathcal{C}}[L_{\bar{U}} A_\alpha] \quad (7.1)$$

on the surviving coarse-curl range after the simultaneous Ward quotient. Either prove annihilation/cancellation or construct a source-faithful completed coefficient that detects the mode.

3. **If strong columns remain incompatible, change the analytic object honestly.** Formulate a range-restricted tube, a direct weak-quotient-dual estimate, or a physically justified curl-free, smoothed, weighted, or separated source class. Do not silently change the intended plaquette observable.
4. **Complete the existing first-jet prerequisites.** Prove (H_{rc}) , the common converted raw/gradient rows, the completed Ward identity, the active-label metric halo when required, and the compatible J -representative atlas.
5. **Only then build the nonzero-source and iterative theory.** Construct source-dependent polymer activities on a common disk, add large fields, prove unit translations and projection, iterate the RG, and begin regulator removal.

7.2 Assumption ledger

Item	What exists	What remains
Selected branch	Finite normalized law, local holomorphy, gauge-covariant push-forward	Equality with the unrestricted raw transform; global chart compatibility
Marked first jet	Exact dual-number routing and conditional fixed-gas convergence at $t = 0$	Common nonzero-source polymer disk and second source jet
Completed support	Literal-union external support and zero J -endpoint halo on aligned cells	Unit-translation theorem and any stronger marked-root metric halo
Affine J tube	Conditional coefficientwise H^∞ rerun and dual bound	(H_J^{conn}) compatible representative atlas
Product U/J tube	Explicit norm and radius under a real-center curvature margin	(H_{rc}) for the full physical family

Item	What exists	What remains
Source kernel	Audited row bounds, exact affine range, exact elliptic factorization	Complete chart/restoration conversion and uniform physical synthesis
Stokes mode	Exact source-normalized finite-model lower bound	Embedding into full multiscale H_1 and Hessian-selected survival
Ward quotient	Exact finite-dimensional quotient/phase theorem	Completed coefficientwise Ward identity and actual uniform quotient norm
Large fields and iteration	Source-free route-level literature context	Source-dependent large-field operation, iteration, projection, running flow
Continuum and infrared	Only a roadmap and conditional reconstruction lemmas	Every construction, nontriviality, OS, infinite-volume, decay, and gap gate

7.3 Clay obligation matrix

Obligation	State at this handoff
Every compact simple group	Open; the calculations are primarily $SU(2)$.
Continuum theory on $\mathbb{R}^4$	Open; only finite regulators appear.
Infinite volume	Open.
Nontriviality	Open.
Local continuum observables and UV behavior	Open.
OS/Wightman-strength axioms and reconstruction	Open.
Finite positive physical mass gap	Open; no infrared bridge exists.
Independent expert review	None recorded.

8 Reproducibility and source boundary

8.1 Repository map

The authoritative machine-readable state is in `STATUS.json`, `CLAIMS.json`, and `audit/objections.json`. The principal research narrative is in `research/LOG.md`. The most relevant notes are grouped below.

Notes/claims	Role
0020–0022	Distinguished-vertex Ursell identity, exact marked Section-2 factorization, and whole-integrand Cauchy.
0023–0024	Literal-union geometry, rooted animals, and explicit ordinary pinned Kotecký–Preiss interface.
0025–0027	Source-faithful conditioning, positive marked resummation, and shifted first-jet synchronization.
0028–0029	Periodic endpoint bridge and completed external support.
0030–0032	Conditional affine/product tubes and support-anchored physical J .
0033–0034	Physical- U quotient synthesis, transported-curl reduction, and physical-width extension.
0035	Diagonal scale-renorming no-go.
0036	Joint Ward quotient, vertical scale trade, and an abstract transverse countermodel.
0037	Exact elliptic factorization plus the source-normalized finite-model coarse-curl Stokes obstruction.

The focused finite-model regression is `tests/test_elliptic_range_factorization_stokes_obstruction.py`. Repository-wide checks are run with

```
python scripts/verify_repository.py
python -m unittest discover -s tests -v
```

The frozen handoff snapshot is checkpoint YM-RG-037 on branch `codex/first-source-audit`, with 141 passing tests and public draft [pull request 5](#). The final immutable commit identifier is recorded in the repository history and pull-request head rather than embedded circularly in this artifact. These commands certify neither the primary literature nor the deep mathematics. They are guards against algebraic drift and status inflation.

8.2 Primary-source audit boundary

The retained local audits visually checked the relevant pages of:

- Balaban’s averaging paper for the linear and nonlinear block-map normalizations [8];
- the gauge-fixing paper for the projected Landau operator [9];
- the variational paper for the three distinct lifts, the background expansion, and the differentiated equations [11];
- *Propagators II* for the multiscale distance/decay transfer [7];
- RG I and II for the selected small-field and cluster scaffolds [12, 13].

Every imported-versus-derived distinction should be preserved. In particular:

- H, H_0, H_1 are distinct operators;
- (6.9) and (6.10) are algebraic consequences, not separately printed displays;
- (6.11) is the derivative consequence of a printed expansion;
- Theorem 6.3 is repository finite-model algebra, not a theorem printed by Balaban and not a proved H_1 embedding.

9 Conclusion

The project did not solve the Yang–Mills existence and mass gap problem. It did produce a useful finite-regulator map of one local-observable bottleneck. The one-mark cluster algebra is substantially organized; exact support and quotient structures are known; and several seductive norm-based repairs have concrete countermodels. The differentiated variational equations have an exact elliptic factorization, while the final Stokes model shows why the normalized range constraints alone should not be mistaken for strong smoothing.

The most important negative discovery is methodological: a Green operator, a weighted norm, a Ward slogan, or a projected Landau condition cannot be read separately from its source normalization, analytic radius, coefficient dual, and domain. The most important positive handoff is a sharp next experiment: determine whether the full multiscale Hessian-selected H_1 range contains the finite-model curl mode, and if it does, test the completed simultaneous U/J weak dual on that range before investing in further strong-column estimates.

A Exact nonclaims

For avoidance of doubt, this paper does not establish any of the following:

1. equality between the selected small-field branch and the unrestricted raw block transform;
2. source-dependent polymer activities on a common nonzero source disk;
3. a common real-center theorem (H_{rc}) for the completed physical family;
4. converted raw/gradient source rows with regulator-uniform constants;
5. the completed coefficientwise simultaneous U/J Ward identity;
6. embedding of the finite Stokes model into the full H_1 range;
7. a source-dependent large-field theorem;
8. RG iteration, marginal projection, or control of the running coupling through a dynamical mass scale;
9. continuum or infinite-volume existence, nontriviality, Euclidean covariance, OS reconstruction, physical-unit exponential decay, or a mass gap;
10. extension from the initial $SU(2)$ calculations to every compact simple group.

B Future-agent falsification checklist

A future agent should reject any proposed continuation that:

- applies the differentiated right-inverse identity to A_c rather than $\kappa_c = \mu_c A_c$;
- identifies H, H_0, H_1 , or calls the nonlinear restoration a gauge transformation;
- reads the inverse in (6.9) as smoothing while deleting its $\Delta_a H_0$ source;
- replaces projected Landau $RD^*K = 0$ by full Landau $D^*K = 0$;
- calls the anisotropic finite periodic model an actual multiscale source patch without proving the embedding;
- takes separate absolute values of the U and induced- J summands before using the Ward quotient;
- relies on cancellation between source columns despite arbitrary source phases;
- spends output decay without checking the same-layer same-cell diagonal;

- hides an $O(\xi^2)$ analytic radius inside a weighted norm constant;
- promotes a finite-regulator, conditional, or model statement to a continuum construction or mass gap.

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